

MUTHUVAIRAVAN K

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47, Manjunatha Layout Main Rd,
Silver Springs Layout,
Munnekollal, Bengaluru,
Karnataka



EDUCATION

SRI MUTHUKUMARAN INSTITUTE OF TECHNOLOGY

Chennai, Tamilnadu
B.E ECE (Apr 2011)

SMJVHSS

Chennai, Tamilnadu
12th (Apr 2007)

Velammal Mat. Hr.Sec.School

Chennai, Tamilnadu
10th (Apr 2005)

Versatile R&D professional with 12 years hands-on experience in Digital and DC-DC power design. Experience in development of micro-controller and Micro Processor based products and solutions. Profound knowledge in hardware development. Proficient with Cadence custom design tools, Altium Designer, Mentor Graphics and Hyperlynx. Experienced in High speed designs such as Ethernet, USB and Flash. Extensive knowledge in circuit debugging techniques, board design and Testing. Hands on experience on the complete hardware development lifecycle.

EXPERIENCE

SENIOR TECHNICAL LEAD

HCLTECH, Bangalore/Mar-2025 – May 2025

- Part of the Verification and validation team.
- Involved in the requirement analysis of the Dialysis machine.

SENIOR DESIGN AND ANALYSIS ENGINEER

Quess Corp Ltd, Bangalore/July 2023 – Nov 2024

Worked for Boeing International, Bangalore

- Part of the Design team in developing the Galley Equipment.
- Designed the Carrier Board the Galley product with interfaces such as RS232, USB, UART, SPI and I2C.
- Experience in design of SOM module with Zync 7000 series FPGA.
- Involved in the complete hardware design lifecycle of the product.
- Designed the Architecture & Hardware Design Document, involved in the Component selection and Designing the Schematics.
- Worked with the mechanical team to perform the Thermal Analysis.
- Performed the Power Integrity for the Carrier Board.
- Hands on experience with tools such as Mentor Graphics and Hyperlynx.
- Hands on experience on Board Bring up activities which involves the complete testing of the product.
- Involved in the interaction with the vendors of the components during the BOM selection and Schematic Design process.

LEAD ENGINEER

Tata Autocomp Systems Ltd, Pune/Aug 2021 – July 2023

Worked for Baxter Innovations and business Solutions Pvt Ltd,Bangalore

- Part of the R&D team in developing the Infusion Pump and Bolus Cord.
- Designed the Front panel PCBA for the infusion pump.
- Involved in the dry run activity of the PCBAs present in the infusion pump.

- Hands on experience in using the Lab instruments such as Oscilloscope, Functional Generator, ESD Gun and IPX chamber.
- Hands on experience in usage of tools such as Altium Designer, Micro Process ALM and Team Center.
- Involved in the testing of ESD, thermal Analysis, IPX and strain relief for the Infusion Pump and Bolus Cord.

HARDWARE DESIGN ENGINEER

Alphion India Pvt Ltd, Chennai, Tamilnadu / Jan 2018 – Aug 2021

- Have designed the schematics for updated version of CTL card microprocessor board using mentor Designer tool.
- Have involved in complete design of the fan tray controller using Pic Microcontroller and ADDA fans. Have supported the pcb team in bringing up the layout design.
- Have undergone certification testing such as EMI, EMC, BUMP, Vibration, QM333 and QM301 for the products which we have made.
- Hands on experience in handling DSO, Logical Analyzers, Variable Power supplies.
- Performed EMI/EMC testing for the Controller Cards and the 1RU unit.

SR.ENGINEER-INDUSTRIAL IOT

Spudweb Technologies Pvt Ltd, Chennai, Tamilnadu / Jul 2017 - Jan 2018

- Involved in the product design of Real Time Monitoring System.
- Have created design specifications and reviews for the Product. Involved in Design, Board bring up and testing of the Product.
- Have lead the team and coordinated with them in completing the product. Have installed the Product at the client Place and have involved in Troubleshooting.

EMBEDDED HARDWARE DESIGN ENGINEER

IOTA Automation India Pvt Ltd, Chennai, Tamilnadu / Jul 2016 - Jul 2017

- Involved in a product development of Liquid Level Indicator.
- Designed the schematic and have involved in Board bring up and testing of the product. Involved in sourcing and purchasing of the components that are used in the circuit.
- Have done the design documentation work and created the user manual for the Product.

RESEARCH ASSOCIATE

EPR Labs, Chennai, Tamilnadu / Dec 2012 - Jun 2016

- Responsibilities: Designing Boards based on various microcontrollers such as 8051,AVR,PIC and ARM.
- Designed and developed the circuit for Hardware and involved in Board bring up and testing. Write software to support new hardware development.
- Research new architecture design to come up with new product designs.
- Have developed the hardware schematic for Raspberry shield and power supply. Developed firmware for this process. Provide management with status update reports.

I hereby declare that the above said information are true to the best of my knowledge.

Place: Chennai

(MUTHUVAIRAVAN.K)

Date :